

Optimizing Broiler Chicken Supply Chains Under Uncertain Average Growth Rate Acceleration

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Abstract—Effective planning of the broiler chicken production chain heavily relies on a comprehensive understanding of the factors influencing growth and final carcass weight. The broiler chicken supply chain includes various stages, from breeders to retailers. This study analyzes the impact of growth uncertainty in broiler chickens by considering aspects of maintenance management, feed consumption, mortality rates, productivity, and market demand scenarios. The research method uses a two-stage stochastic program and Mixed Integer Linear Programming (MILP) to optimize lot-sizing and lot-allocation in the broiler chicken supply chain planning. MILP is applied to maximize the efficiency and responsiveness of lot-sizing and lot-allocation to market demand. Two-stage stochastic modeling allows for the separation of long-term and short-term decisions, enabling adjustments based on uncertain conditions. By simulating various production scenarios, potential issues and improvement opportunities within the supply chain can be identified. Production data is analyzed to ensure that the size and weight of broiler chickens meet market specifications and demands, thereby enhancing customer satisfaction and operational efficiency. The implementation of an integrated broiler chicken supply chain is believed to improve production and distribution efficiency, enabling the fulfillment of market demand at optimal costs. By ensuring this alignment, the research contributes to increased customer satisfaction and operational efficiency. Meeting market specifications not only improves product quality but also strengthens the company's competitive position in the market. Higher operational efficiency is also achieved through more effective reductions in production and distribution costs.

Keywords—growth uncertainty, inventory, lot-allocation, lot-sizing, supply chain.

I. INTRODUCTION

Indonesia, with a population of 281,603.8 thousand people [1], has a high demand for food, especially protein. Adequate food and nutrition are an important investment in building quality human resources and are a prerequisite for fulfilling basic rights such as education and work. One of the animal protein sources widely consumed in Indonesia is broiler chicken. Protein and amino acids play increasingly complex roles in regulating various aspects of bodily health. Those matter contribute to regulating body composition, bone health, digestive function and bacterial flora, glucose homeostasis, cell signaling, and satiety [2]. Broiler chicken meat has a good taste and high nutritional content, making it a popular choice among the community. In addition, broiler chicken farming is of interest to the community both for consumption and as a large-scale and small-scale business. The high demand for

broiler chicken has driven the broiler chicken industry to continue to grow [3].

The broiler chicken industry faces several challenges. Growth and carcass weight are influenced by genetics, nutrition, management practices, environment, animal health, and individual variation [4]. This variability can cause production fluctuations, complicating planning and decision-making [5]. The complex supply chain, involving farmers to retailers, faces uncertainties in demand, supply, and price, disrupting efficiency and increasing costs [6]. Climate change also impacts growth, leading to decreased production and quality [6]. Closed-system poultry houses regulate air and temperature, mitigating weather effects, unlike open systems which need intensive management [7]. Broiler products include whole and carcass chickens, marketed whole or in parts [8]. Efficient supply chains are crucial for meeting customer demand, enhancing operational performance, and gaining competitive advantages [9,10]. The broiler supply chain involves breeders, hatcheries, farms, slaughterhouses, wholesalers, and retailers, each playing a vital role in production and distribution [11].

Several previous studies have been conducted to address the challenges in the broiler chicken industry. Broiler chicken production aims to optimize energy use and chicken quality. Mixed integer linear programming (MILP) is one of mathematical modeling methods that minimize energy consumption and maximize chicken quality. Some research also show that the stochastic model is effective in fulfill specific constraints and offers the lowest feed cost [15]. The technical efficiency of poultry production in Peninsular Malaysia was analyzed using a stochastic frontier production function. The research results indicate that technical progress in the early period was rather slow but improved during the second period due to structural and technological changes aimed at modernizing the broiler chicken industry [16]. Growth is closely related to health and strain, contributing significantly to carcass health and performance [17]. Climate change also significantly impacts the growth of broiler chickens, particularly through extreme conditions such as water scarcity. This scarcity can lead to a 20% reduction in water consumption, thereby affecting feed intake [18]. Using two-stage stochastic linear programming for production planning in the industry involves considering scenarios of mortality rates, productivity, and sales prices [19]. An ordering model with deteriorating and growing item conditions affects the demand level and selling price of a product, proposed a linear programming model for broiler chicken supply chain integration [20].

This research is expected to make a significant contribution to the broiler chicken industry in Indonesia and help improve national food security. The broiler chicken industry in Indonesia is facing several challenges, such as uncertainty in growth and carcass weight, suboptimal supply chain efficiency, and the impact of climate change. This research aims to develop a two-stage stochastic optimization and MILP model to optimize broiler chicken supply chain planning. This model is expected to increase supply chain efficiency, reduce production costs, improve product quality, and enhance food security. The research is expected to make a significant contribution to the broiler chicken industry in Indonesia and help improve national food security.

II. MATERIALS AND METHODS

A. System Description

Before operation, the farm and hatchery agree on an order plan for the planning horizon. Daily DOC (day-old chicks) arrivals are adjusted based on farm capacity. Feed rations are arising and is performed repeatedly [12].

The just-in-time (JIT) method will be compared, which has gained significant interest due to its benefits such as cost reduction, lower inventory levels, improved product quality, shorter processing times, enhanced responsiveness, higher customer satisfaction, and competitive advantage. JIT emphasizes timely arrival of materials in the right quantities and locations. Companies depend heavily on suppliers for timely deliveries, minimizing downstream inventory. To avoid stockouts or losses, accurate demand forecasting and rapid response to market changes are crucial [13]. An optimal order quantity model for items with exponential deterioration should consider opportunity costs based on compound interest and backordering, aiming for an accurate, nearly optimal, and easily understandable closed-form solution [14].

B. Model

Descriptive statistical analysis is used as a suitable data analysis method to describe the growth of broiler chickens and ordering costs. To determine the optimal solution for the case of broiler chicken growth and ordering costs, the assumptions used in this study are as follows:

1. Orders are made once and reconfirmed when the order is ready to be shipped to the farm.
2. The orders include DOC (Days-Old-Chick) or chicks, livestock feed (MT), and medicines, vaccines, and chemicals (OVK).
3. At the beginning of the period, there is an inventory of broiler chickens derived from the extrapolation of inventory data from the previous period, documented in the chicken stock report of last period.
4. Adjustments (additions or reductions) are made before the order is shipped to the farm based on the stock adjustments.
5. The maintenance period is based on the average maintenance age, which is 5 weeks over 1 year.
6. The standard maintenance weight is based on the average maintenance weight after harvest over 1 year.
7. There is a maximum and minimum number of chickens that must be met according to market demand, adjusted by the number of Chick In and Chick Out over 1 year.
8. There is damage or mismatch with the demand during the maintenance period, taken from the average depletion value over 1 year.

9. Chickens that have reached time J are taken whole without considering cleaning and cutting processes.

Before the operation, the farm and hatchery agree on an upcoming order plan for the planning horizon. Each day within the planning horizon, the number of DOCs (Day-Old Chicks) arriving at the farm is adjusted based on the farm's capacity. The feed ration for chicken consumption during the period is submitted 3 months before the rearing period and readjusted 1 week before the planned arrival of DOCs at the farm. Subsequent feed orders are placed during the ongoing period, adjusting to the chicken's needs to anticipate feed surplus or shortage during the rearing period. There are no additional DOCs added during the rearing period. Chicken reduction at the farm occurs due to chicken orders from the slaughterhouse and mortality, as natural death and death due to disease. Growth uncertainty during rearing, like disease or hindered growth, results in defective products. The slaughterhouse produces carcass/whole chicken for market and restaurants.

The challenges above need to be addressed to ensure the sustainability and food security of Indonesia. This research aims to develop a two-stage stochastic optimization and MILP model to optimize broiler chicken supply chain planning. This model is expected to increase supply chain efficiency, reduce production costs, improve product quality, and improve food security. The Monte Carlo method is also used in this research. The Monte Carlo method is a statistical sampling technique useful for estimating accurate solutions to quantitative problems. This method randomly calculates the values of uncertain variables that arise and is performed repeatedly [12].

Order changes are considered non-applicable; once orders are placed, they are confirmed when DOCs are ready to be shipped to the farm. A model is proposed to solve the lot-allocation problem under broiler growth uncertainty, optimizing lot allocation on farms while considering chicken product demand. The model determines lot allocation based on chicken age and feed consumption. MILP is used to accommodate growth and feed consumption uncertainty through various growth scenarios.

The broiler chicken lot-sizing model based

$$\min_{z_{sizing}} = \sum_{t \in T} C^0 x_t + \sum_{t \in T} \sum_{\omega \in \Omega} \sum_{p \in P} \rho^\omega [C^0 y_{tp} + C_p^H I_{tp}^\omega + C_p^F F_{tp}^\omega +] + C_{tp}^P P_{tp} \quad (1)$$

Constraints (2a) and (2b) ensure that the order size in each week is not less than the minimum order quantity and not more than the maximum order quantity that has been set

$$x_t \geq Q^{min} \quad \forall t = 1, \dots, (|T| - J + 1); \quad (2a)$$

$$x_t \leq Q^{max} \quad \forall t = 1, \dots, (|T| - J + 1); \quad (2b)$$

Constraint (3a) regulates the distribution of mature chickens from unadjusted orders to the production line for processing into final products. Constraint (3b) accounts for reactive decisions to modify orders according to the conditions and changes that occur in each scenario.

$$\sum_{g \in G|t-J-g+1 \geq 1} B_g^\omega x_{t-J-g+1} = \sum_{l \in L} z_{tl}^\omega, \forall t = (J + \min G), \dots, (2J + L + \min G - 1), \omega \in \Omega \quad (3a)$$

$$\sum_{g \in G} B_g^\omega x_{t-J-g+1} \sum_{g \in G|t-J-g+1 \geq 1} B_g^\omega q_{t-J-L-g+1} = \sum_{l \in L} z_{tl}^\omega, \forall t = (2J + L + \min G - 1), \dots, |T|, \omega \in \Omega \quad (3b)$$

Since the data is random, we need to perform normalization to ensure that the data has a uniform scale or is

consistent with a certain range. Therefore, the equation for the inventory control model is as follows:

$$I_{tp}^{\omega} = I_{tp}^0 + R_p z_{tl}^{\omega} + F_{tp} - D_{tp}, \forall t = J + \min G, l \in L, p \in P, \omega \in S \quad (4a)$$

$$I_{tp}^{\omega} = I_{t-1,p}^{\omega} + R_p z_{tl}^{\omega} + F_{tp} - D_{tp}, \forall t = (J + \min G + 1), \dots, |T|, l \in L, p \in P, \omega \in S \quad (4b)$$

$$I_{tp}^{\omega} \geq I_{tp}^{\min}, \forall t = (J + \min G + 1), \dots, |T|, l \in L, p \in P \quad (4c)$$

Constraints (4a), (4b), and (4c) ensure that the inventory of broiler chicken products is calculated correctly at the end of each time period, considering the initial inventory, new production from slaughtered chickens, and demand for broiler chicken products. Constraints (5a) and (5b) set the minimum limits for the inventory of broiler chicken products that must be maintained at all times.

$$x_t \geq 0, \forall t = 1, \dots, (|T| - J - \min G + 1); \quad (5a)$$

$$z_{tl} \geq 0, \forall t = (J + \min G), l \in L, \omega \in S; \quad (5b)$$

$$I_{tp}, F_{tp} \geq 0, \forall t = (J + \min G), p \in P, \omega \in S; \quad (5c)$$

Constraints (6) to (11) explain the nature and limitations of the decision variables used in this model. Summary of the notation used in the broiler chicken lot-sizing model under growth uncertain.

Set:

- T : Set of weeks in the time horizon (ordered set)
- G : Set of deviations from the standard maturity age of chickens
- S : Set of chicken growth scenarios
- L : Set of production lines
- P : Subsets of chicken products in line $l \in L$

Parameter:

- C' : Unitary cost of day-old-chickens [Rp]
- Q : Minimum weekly order quantity of day-old chickens [chickens]
- $\bar{Q}$: Maximum weekly order quantity of day-old chickens [chickens]
- J : Standard number of weeks for an old-day chicken to reach weight [weeks]
- B'_g : Percentage of old-chickens that reach target weight in $J + g$ weeks under scenario $\omega \in S$ [dimensionless]
- P' : Probability of scenarios $\omega \in S$ [dimensionless]
- R_l : Number of units of chicken product $p \in P$ obtained from whole chicken [unit/chicken]
- I_p^0 : Inventory of chicken product $p \in P$ on week $t \in T$ [units]
- D_l : Demand of chicken product $p \in P$ on week $t \in T$ [units]
- C'_i : Unitary cost of holding product $p \in P$ in inventory [Rp/chicken]
- C''_i : Unitary cost of unsatisfied demand of product $p \in P$ [Rp/chicken]
- C_i : Unitary cost of produced of product $p \in P$ in inventory [Rp/chicken]

Variables

First-stage decision variables

- x_t : Size of day-old chicken that arrives on week $t \in T$ [chickens]

Second-stage decision variables

- y_{tp} : Order modification of day-old chicken that arrives on week $t + L \in T$ [chickens]

- z_{tl}^{ω} : Number of chickens to send to production line $l \in L$ under the scenario $\omega \in S$ [unit/chicken]
- I_{tp}^{ω} : Inventory of chicken product $p \in P$ on week $t \in T$ under the scenario $\omega \in S$ [chickens]
- F_{tp}^{ω} : Number of unsatisfied products to production line $l \in L$ under the scenario $\omega \in S$ [chicken]

The goal is to minimize production costs and feed consumption based on various growth scenarios.

$$\min_{z_{allocation}} = \sum_{t \in T} \sum_{h \in H} \sum_{j \in J} C_{th}^f x_{thj} \quad (6)$$

Equation (6) ensures that the order size in a given week matches the model solution of equation (1). Since the last lot under consideration must be ready for slaughter by the end of the planning period, we can only order up to period $|T| - |J| + 1$. Furthermore, the usage will be adjusted according to the growth of the chicken with the following constraints:

$$x_{1hj} = x_{hj}^0, h \in H^0; j \in J | x_{hj}^0 > 0; \quad (7a)$$

$$x_{thj} \leq 0, t = 1, \dots, |J| - 1; h \in H; j = t + 1, \dots, |J| - 1; \quad (7b)$$

Equation (7b) reflects the model assumption that takes into account chickens that are already in the rearing cage at the beginning of the planning period. Equation (7a) and (7b), on the other hand, serves as a constraint to avoid placing chickens with age $j > t$ where j is the age of the chicken and t is the planning period) into cages that are still in the preparation stage at the beginning of the planning period. Then, modelling the growth cycle of broiler chickens is modelled by the following equation:

$$x_{thj} = x_{t+1,j+1}, t = 1, \dots, |J| - 1; h \in H; j = t + 1, \dots, |J| - 1; \quad (8a)$$

$$x_{th1} = 0, t = |T| - |J| + 2, \dots, |T|; h \in H; \quad (8b)$$

Equation (8a) and (8b) ensures that each chicken housed in a cage will grow to reach slaughter weight at maturity (at age $|J|$). It serves as a constraint to prevent the arrival of new chickens to the farm in the final weeks of the planning period, as those chickens will not have enough time to reach maturity and slaughter weight before the end of the period.

With the decision variables as follows:

$$x_{thj} \in \mathbb{Z}_+^1, t \in T; h \in H; j \in J; \quad (9)$$

Summary of the notation used in the broiler chicken lot-sizing model under growth uncertain

Sets

- T : Set of weeks in the time horizon (ordered set)
- J : Set of the standard maturity age of chickens [weeks]
- H : Set of chicken growth houses [units]

Parameter

- C'_i : Unitary cost of fattening of chicken $p \in P$ in inventory [kg/chicken]

Variables

First-stage decision variables

- x_{thj} : of feed used during the period

Second-stage decision variables

- y_{tp} : Order modification of day-old chicken that arrives on week $t + L \in T$ [chickens]

- z_{tl}^ω : Number of chickens to send to production line $l \in L$ under the scenario $\omega \in S$ [unit/chicken]
 I_{tp}^ω : Inventory of chicken product $p \in P$ on week $t \in T$ under the scenario $\omega \in S$ [chickens]
 F_{tp}^ω : Number of unsatisfied products to production line $l \in L$ under the scenario $\omega \in S$ [chicken]

III. RESULT AND DISCUSSION

A. Inventory Control Model

Based on the available data, data selection has been carried out by analyzing the probability function and showing that the data is normally distributed with a growth rate of 90%. Thus, the growth scenario selected in this study consists of 1 growth scenario. Chickens that reach the target weight at the standard age week with a 90% life rate with sales demand according to demand variations. This scenario is taken based on the results of calculations and data analysis in 1 year.

The inventory control model conducted to ensure broiler inventory always meets demand and does not exceed storage capacity so that companies can improve their supply chain efficiency, reduce operational costs, and increase customer satisfaction through better inventory management. By ensuring all broiler lot-sizing values meet the minimum capacity requirements in equations (6) and (7) with constraints (8) and (9), then given an inventory based on two production lines, the number of chickens to be put into production line (1).

(i) which is chicken in the form of carcasses with the assumption that the chicken is channeled to the production line as a whole and then will be cleaned and slaughtered at the slaughterhouse (not done in the cage) and the production line (ii) which is a good whole chicken and produced as a whole (alive) using the inventory model in equation (4b) as the optimal inventory control values for JIT and Monte Carlo simulation are as follows:

TABLE I. OPTIMAL SOLUTION FOR PRODUCTION RATE AND ORDERING INVENTORY MODEL ACCORDING TO JIT AND MONTE CARLO SIMULATION

(T)	(P ¹)	x_t (in million)		I_{tp}^ω (in million)	
		JIT	Monte Carlo	JIT	Monte Carlo
1	0.9	50	85.5	186.3	220.9
⋮	⋮	⋮	⋮	⋮	⋮
50	0.9	50	125.2	104.5	91.497
51	0.9	50	124.1	105.1	110.4

Graphically, the results are shown in Fig. 1 and Fig. 2.

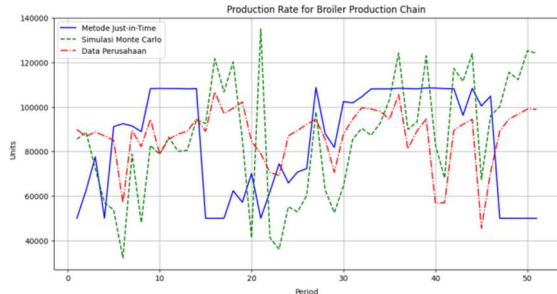


Fig. 1. Optimal solution of production rate for ordering according to just-in-time, monte carlo simulation and actual data

This graph illustrates the optimal production level for the company for 51 weeks. Fig. 1 shows a comparison graph between the number of production units produced using the just-in-time, monte Carlo simulation and actual data. Each method has different characteristics and implications

regarding stability, flexibility, and responsiveness to market demand changes. The Just-in-Time (JIT) method aims to minimize inventory by ordering only when needed, based on actual demand. JIT shows a more stable ordering pattern with fewer fluctuations. There are some sharp drops, particularly in the 20th week and the last week, with orders generally ranging from 60,000 to 100,000 chickens.

Monte Carlo simulation shows high variability, reflecting the ability to adapt to various unexpected demand scenarios. Significant fluctuations are observed, with orders reaching over 120,000 at their peak and dropping to around 40,000 in some weeks. The Monte Carlo simulation provides a more realistic view of various possible scenarios, aiding in more flexible planning and responsiveness to demand changes.

Company data reflect the actual ordering patterns executed by the company over the same period. The chart shows that the ordering pattern exhibits moderate fluctuations, with quantities ranging from 60,000 to 100,000 chickens, reflecting a combination of stability and flexibility approaches.

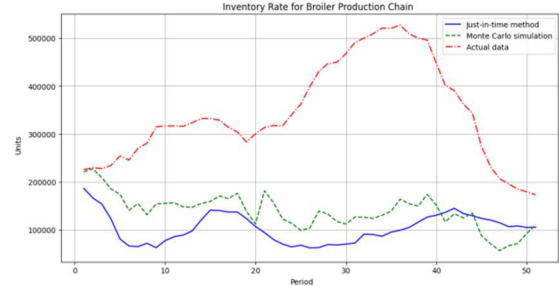


Fig. 2. Inventory Rate according to just-in-time, monte carlo simulation and actual data

Fig. 2 illustrates the optimal inventory levels for the company over 51 weeks by comparing inventory levels between the JIT method, Monte Carlo Simulation, and Company Data. The chart shows that the JIT method maintains relatively low inventory levels, with the number of broiler chickens ranging between 50,000 and 150,000. It experiences fewer fluctuations compared to other methods, aiming to reduce storage costs and the risk of spoilage. However, the initial ordering time and maintenance period are very long, making this method prone to stockouts if there is a sudden demand spike.

The Monte Carlo simulation graph shows that this method maintains higher inventory levels than JIT but lower than company data, with quantities ranging from 100,000 to 200,000. It requires larger storage capacity and higher costs compared to JIT. Although this method shows greater variability than JIT, it is still quite controlled. The higher flexibility aims to anticipate changes in demand.

Company data indicates that the company maintains fairly high inventory levels, ranging from 200,000 to 500,000 chickens, peaking around weeks 35 to 45. There are significant fluctuations in inventory levels, reflecting the company's adaptation to market demand and actual production conditions. However, high initial ordering levels can lead to high storage costs and a greater risk of spoilage.

During periods of anticipated increased demand, the company opts to build a buffer by maintaining higher inventory levels. Generally, inventory levels rise from weeks 1 to 14, indicating potential demand increases as the company prepares by storing more inventory to meet rising needs. Levels then decline from weeks 15 to 44, indicating potential demand stabilization or decline as the company reduces

inventory after a high-demand period. Inventory levels do not reach zero during weeks 48-51, suggesting overproduction and anticipation of increased future demand (end of year). This indicates a clear demand trend during this period.

From the calculations presented in the table, the scenario used shows a growth percentage that meets the target body weight of 90%. This indicates a high percentage of chickens reaching the target weight, suggesting good rearing conditions and low mortality rates. A growth rate not meeting the target weight of 10% also shows management efficiency, with minimal losses during rearing. The percentage of chicken addition due to demand adjustments is 2% per period, indicating stable population growth, giving the company time to adjust production and inventory. The first scenario shows a cautious production and inventory strategy. The company maintains high inventory levels at the beginning and end of the period, possibly to avoid stockouts due to demand fluctuations. Production tends to be high (almost always above 36,000 chickens), indicating both methods focus on demand fulfillment.

In addition to meeting demand, the company also significantly maintains low inventory levels and stable production fluctuations, reflecting efforts to minimize inventory costs and adapt to spoilage levels, thus making production costs more efficient and allowing production control to adjust to demand levels. An optimal production and inventory strategy focuses on demand fulfillment, minimizing inventory costs, and adapting to spoilage levels. This strategy results in stable chicken population growth and enables the company to adjust production and inventory effectively.

B. Model Lot-Sizing for Broiler Chicken Growth

After calculating the optimal solution for the Inventory model, we will then compute the ordering costs according to the Lot-Sizing model for broiler chicken growth as per equation (1) with ordering constraints (2a) and (2b). Subsequently, chickens entering the production line are counted based on reaching age J according to constraints (3a) and (3b) using decision variables (5a), (5b), and (5c), thereby obtaining the optimal solution for the Lot-Sizing model of broiler chicken growth in the just-in-time method and Monte Carlo simulation as follows:

TABLE II. OPTIMAL ORDERING COST SOLUTION FOR THE LOT-SIZING MODEL OF BROILER CHICKEN GROWTH USING JIT METHOD AND MONTE CARLO SIMULATION

(T)	(P ¹)	min z_{sizing}^1 (in million)	
		JIT	Monte Carlo
1	0.9	1,512,450.9	2,747,842.2
⋮	⋮	⋮	⋮
50	0.9	1,002,206.7	3,132,709.7
51	0.9	1,007,512.7	3,016,635.2
Total Cost		63,856,645.7	187,605,888.2

Graphically, the results are shown in Fig. 3 as follows:

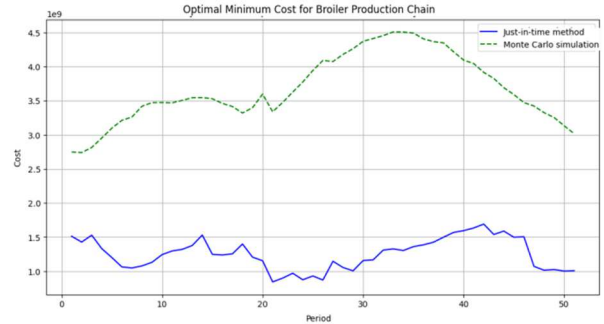


Fig. 3. Optimal production cost solution for ordering according to just-in-time and Monte Carlo simulation

Fig. 3 graph illustrates how optimal production and inventory levels impact the costs required for lot-sizing the growth of broiler chickens due to the influence of storage costs over 51 periods for the company. The optimal cost solution shows a dynamic pattern, reflected in the fluctuations of production levels throughout the planning horizon. The total cost in the scenario using the Monte Carlo simulation tends to be higher compared to the just-in-time scenario. This is due to the high inventory costs, which result in greater storage costs and higher production costs because the high production strategy involves higher production costs, especially due to storage, production, and ordering costs.

The total cost in just-in-time tends to be lower compared to the Monte Carlo simulation. This is because of lower inventory costs as lower inventory levels mean lower storage costs and the production efficiency practiced by the company by implementing a more efficient production strategy to tackle higher spoilage rates. This indicates that the company may have achieved a relatively good level of efficiency in management and production, but the higher costs may be due to the need to address high spoilage rates.

C. Broiler Growth Lot-Allocation Model

After calculating the optimal solution to minimize production costs and feed consumption based on various scenarios, calculations will then be made for ordering costs according to the Lot-Allocation model according to the growth of broilers according to equation (12) with ordering restrictions according to (13). Then, feed usage will be calculated based on the constraints for fattening (16) and (17) calculated based on the number of chickens reaching age J according to the constraints (14) and (15) in accordance with the decision variables (18), so as to obtain the optimal solution for the Lot-Allocation model, so that the optimal solution for the Lot-Allocation model of broiler growth in the just-in-time method and Monte Carlo simulation is obtained as follows:

TABLE III. OPTIMAL SOLUTION OF FATTENING COST OF LOT-ALLOCATION MODEL OF BROILER GROWTH USING JIT METHOD AND MONTE CARLO SIMULATION

(T)	min $z_{allocation}^1$ (in million)	
	Just-in-time	Monte Carlo
1	1,485,000.0	2,539,736.1
⋮	⋮	⋮
50	1,816,650.0	4,549,690.9
51	1,811,700.0	4,495,226.3
Total Cost	139,457,188.2	143,738,225.3

Graphically, the results are shown in Fig 4 as follows:

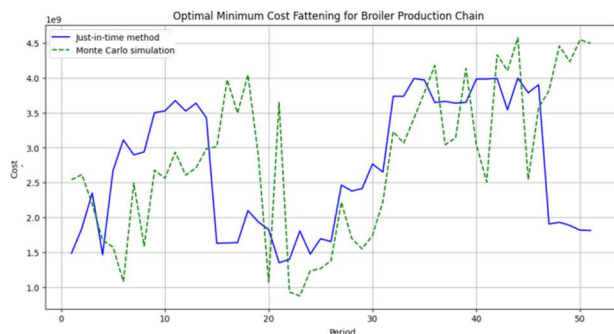


Fig. 4. Optimal solution of fattening cost for ordering according to just-in-time and Monte Carlo simulation

Fig. 4 illustrates that feed costs per period tend to be higher in the Monte Carlo simulation compared to just-in-time. This is due to the more conservative production and inventory strategy in the Monte Carlo simulation, where the company maintains higher inventory levels. Overall, the total feed cost per period decreases from the Monte Carlo simulation. This indicates that as production costs stabilize and spoilage rates increase (represented by the third scenario), the company tends to implement a more efficient feed usage strategy, possibly through a more aggressive production and inventory strategy.

The pattern of feed cost fluctuations per period is relatively consistent in both the just-in-time method and the Monte Carlo simulation. This suggests that other factors may influence feed costs besides the survival rate and spoilage level, such as feed prices, the type of feed used, and cage efficiency.

IV. CONCLUSION

The two-stage stochastic model and MILP have proven effective in separating long-term and short-term decisions in the broiler chicken production. This allows for quicker adjustments to uncertain conditions in the broiler chicken production chain, both for whole chickens and carcasses. Simulations of various production scenarios successfully identified potential problems in the broiler chicken production chain, such as production imbalances leading to oversupply. Regular analysis of historical data and demand projections adjusts the broiler chicken production capacity during the ongoing period. The implementation of the stochastic model and production scenario simulations can strengthen the broiler chicken production chain to be more adaptive to changes in demand in uncertain market conditions. Production planning strategy recommendations include attention to production capacity adjustments, analysis of historical initial order data, demand projections, and the implementation of product quality control monitoring technology.

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